

TANGO: Temporal-Spatial Co-Scheduling for Multi-SLO LLM Serving

Abstract—Online Large Language Model (LLM) serving faces the challenge of delivering high throughput while satisfying heterogeneous latency Service-Level Objectives (SLOs). Each request undergoes a prefill phase to produce the first token, followed by a decode phase that generates subsequent tokens, giving rise to two latency metrics: time-to-first-token (TTFT) and time-per-output-token (TPOT). However, existing LLM serving systems struggle to adapt resource allocation to the varying urgency of requests across prefill and decode phases under heterogeneous SLOs. We therefore propose TANGO, a temporal-spatial co-scheduling framework for Multi-SLO LLM serving. TANGO comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. The dispatcher prioritizes requests based on phase-aware SLO deadlines, while the token allocator regulates per-step workloads using predicted execution time. Together, they coordinate request ordering and batch composition to improve first-token responsiveness and inter-token timeliness. Experimental results show that TANGO improves overall SLO attainment by **58.72%**, and achieves **1.52×** higher goodput on average than state-of-the-art systems.

I. INTRODUCTION

Online large language model (LLM) services consolidate requests from multiple applications on shared model instances to improve GPU utilization and reduce serving costs. These applications impose distinct latency requirements: some requests are sensitive to the delay before the first output token, as in interactive chat, whereas others are sensitive to the delay between consecutive output tokens, as in real-time translation. These latency requirements are associated with two phases of LLM inference: during *prefill*, the model processes the prompt and generates the first output token; during autoregressive *decode*, it generates subsequent tokens one at a time. Time to first token (TTFT) measures the delay from request arrival until the first output token is returned, while time per output token (TPOT) measures the average delay per subsequent token. Online serving systems must therefore maintain high throughput while meeting application-specific TTFT and TPOT service-level objectives (SLOs).

Facing such a scenario, LLM serving aims to improve goodput by satisfying both TTFT and TPOT SLOs for requests with heterogeneous latency requirements. To support this goal, serving systems combine continuous batching, which dynamically admits new requests into the active batch between iterations, with chunked prefill, which divides long prompts into smaller segments that can be processed alongside decoding requests. This combination improves GPU utilization and serving efficiency and is adopted by state-of-the-art inference engines such as vLLM and SGLang.

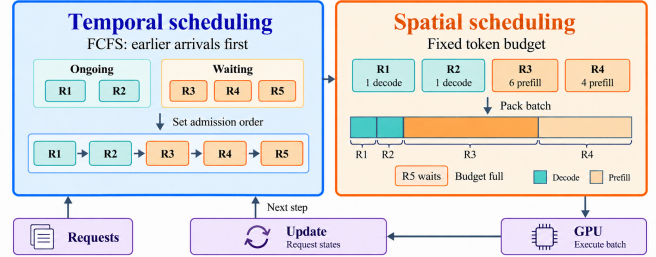


Fig. 1. Temporal and spatial scheduling in LLM serving.

As shown in Fig. 1, incoming requests first enter the engine’s waiting queue. At each execution step, the scheduler selects waiting requests to run alongside ongoing requests. It then allocates the step’s token budget among these requests to form a batch, specifying the sizes of prefill chunks and the number of decode tokens to include. The engine then executes the batch, updates request states, and repeats this process for the next step. We abstract this process into two coupled scheduling dimensions: *temporal scheduling*, which determines request execution priority and admission, and *spatial scheduling*, which determines token allocation within each batch. These two decisions jointly shape request queueing delay and step duration, thereby affecting TTFT and TPOT.

Existing throughput-oriented engines commonly use first-come-first-served (FCFS) admission for temporal scheduling and a fixed maximum token budget for spatial scheduling, without adapting these policies to the heterogeneous SLOs of individual requests. To assess the impact of these policies, we evaluate vLLM’s default FCFS scheduler on Qwen3-14B using the mathematical-reasoning workload with heterogeneous TTFT and TPOT requirements. Among requests with tight TTFT SLOs, only 48.89% satisfy both SLOs, with TTFT violations accounting for most failures. For requests with tight TPOT SLOs, joint SLO attainment is similarly low at 12.00%, with TPOT violations as the main cause. Across the workload, only 53.48% of requests satisfy both TTFT and TPOT SLOs. This low joint attainment highlights the limitations of SLO-unaware scheduling under heterogeneous latency requirements.

First, SLO-unaware temporal scheduling leads to an admission order that does not reflect request urgency. Under FCFS, a request with a tight TTFT objective can wait behind earlier requests with ample latency slack. The urgent request may exhaust its TTFT budget in the queue, even though the earlier

requests could tolerate being served later. FCFS therefore misses the opportunity to defer less urgent requests and use their slack to reduce queueing delays for requests with tighter SLOs.

Second, existing spatial scheduling policies allocate tokens under a fixed per-step budget without considering the TPOT requirements of active decode requests. Because decoding requests must wait for the entire batch to finish before receiving their next tokens, admitting more prefill work can prolong their inter-token delays. With 2,048 decode tokens fixed, increasing prefill work from zero to 2,048 tokens raises mean step time from 329 ms to 759 ms. Moreover, the same total token count can yield different step times: under a 4,096-token budget, pure-prefill and pure-decode steps take 611 ms and 683 ms, respectively, whereas an evenly mixed step takes 759 ms. A fixed token budget therefore cannot reliably bound step duration or adapt prefill allocation to the latency tolerance of decoding requests.

Addressing these inefficiencies requires coordinating request-level urgency with batch-level execution time. First, temporal scheduling involves deciding which waiting requests to admit and which ongoing requests can be deferred under resource contention. These requests are governed by different active SLOs and have consumed different portions of their latency budgets, so determining their execution order hinges on a consistent comparison of urgency across phases. Second, spatial scheduling couples the latency requirements of prefill and decode requests through token allocation within a shared batch. Allocating more tokens to prefill can advance requests with urgent TTFT objectives but also prolong the wait for decoding requests, whereas overly restricting prefill can leave those urgent requests waiting. Balancing these competing demands requires accurately estimating step time under varying prefill/decode compositions and execution contexts, then using that estimate to guide per-step token allocation.

We therefore propose TANGO, a multi-SLO LLM serving system that jointly controls temporal request dispatch and spatial token allocation. TANGO aims to improve goodput while satisfying heterogeneous TTFT and TPOT requirements. It comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. The deadline-aware request dispatcher assigns phase-aware deadlines according to each request’s active SLO and uses them to guide admission and preemption, prioritizing requests with tighter latency budgets. The step-time-model-guided token allocator uses a step-time model to predict the execution time of mixed prefill/decode batches and adjusts their prefill token budgets according to the remaining latency slack of decoding requests.

This paper makes three major contributions:

- **Investigating scheduling inefficiencies in multi-SLO LLM serving.** The analysis motivates us to account for heterogeneous request urgency and mixed prefill/decode interference through joint control of request dispatch and token allocation.
- **The design of a deadline-aware request dispatcher that prioritizes requests across inference phases.** By

TABLE I
COMPARISON OF TANGO WITH REPRESENTATIVE SERVING SYSTEMS.

	Continuous Batching	Heterogeneous SLOs	Runtime Reprioritization	Predictive Token Budgeting
vLLM	✓			
SOLA	✓		✓	
JITServe	✓	✓	✓	
TANGO	✓	✓	✓	✓

mapping each request’s active SLO to a comparable deadline, it guides request admission and preemption, mitigating temporal inefficiency caused by SLO-unaware queueing.

- **The design of a step-time-model-guided token allocator that adapts batch composition to latency constraints.** Based on predicted step time, it converts the remaining TPOT slack into a prefill token budget, mitigating mixed prefill/decode interference while preserving progress for requests with tight TTFT requirements.

We evaluate TANGO against vLLM, SOLA, and JITServe using Llama-3.1-8B, Qwen3-14B, and Qwen3-32B under chatbot, mathematical-reasoning, and software-agent workloads. Compared with state-of-the-art systems, TANGO improves overall SLO attainment by 58.72% and achieves $1.52\times$ the goodput on average.

II. RELATED WORK

LLM Serving Systems. Recent studies have developed efficient LLM serving systems that improve inference throughput and latency through scheduling and memory optimizations. Orca introduced iteration-level scheduling, allowing requests to enter and leave a batch between decoding iterations to improve both utilization and responsiveness. vLLM further improved serving throughput through PagedAttention, which reduces KV-cache fragmentation and enables flexible memory sharing across requests. SGLang exploits prefix sharing and structured execution to accelerate multi-call LLM workloads. These systems establish continuous batching and efficient memory management as the foundation of modern LLM serving. However, throughput-oriented schedulers commonly use an SLO-agnostic request order and a fixed upper bound on the number of tokens processed in each iteration, without adapting either decision to the heterogeneous TTFT and TPOT requirements of individual requests.

Prefill–Decode Scheduling. The prefill and decode phases exhibit distinct execution characteristics: prefill processes many prompt tokens in parallel, whereas decode repeatedly processes one token per request. Mixing the two phases improves GPU utilization but can prolong an iteration and delay the next tokens of ongoing requests. Sarathi-Serve mitigates this interference through chunked prefill and stall-free batching, using decode-maximal batches to improve serving capacity under tail-latency constraints. DistServe instead places prefill and decode on different GPUs and separately optimizes their resource allocation and parallelism under TTFT and TPOT constraints. These approaches regulate interference through configured chunk sizes, phase-level batching policies,

or resource separation. They do not directly translate the remaining latency slack of active requests into an iteration-specific token budget for a colocated mixed batch.

SLO-Aware LLM Serving. Recent systems explicitly optimize the fraction of requests satisfying latency SLOs rather than raw throughput alone. SOLA formulates a fine-grained scheduling space that adjusts both request execution order and iteration workload according to request and system states, thereby balancing TTFT and TPOT. JITServe estimates uncertain request characteristics and allocates each request sufficient serving bandwidth while grouping compatible requests to improve batch efficiency and goodput. However, existing systems do not adjust the per-step token allocation within a mixed batch according to the request-specific TTFT and TPOT slack. Table I compares TANGO with representative serving systems.

III. BACKGROUND AND MOTIVATION

A. Multi-SLO LLM Serving

LLM Inference and Latency SLOs. An online LLM serving engine processes each request in two phases. During *prefill*, the engine processes all prompt tokens in parallel, constructs their key-value (KV) caches, and produces the first output token. During *decode*, it reuses the KV caches and autoregressively generates subsequent output tokens, typically one token per request in each iteration. The two phases expose different forms of latency and corresponding service-level objectives (SLOs): a request must first wait for its prompt to be processed, and must then make timely progress over a sequence of decoding iterations. Let request r_i arrive at time a_i and generate n_i output tokens, where $t_{i,j}$ denotes the time at which its j -th output token is returned. Its time to first token (TTFT) and time per output token (TPOT) are defined as

$$\begin{aligned} \text{TTFT}_i &= t_{i,1} - a_i, \\ \text{TPOT}_i &= \frac{1}{n_i - 1} \sum_{j=2}^{n_i} (t_{i,j} - t_{i,j-1}), \quad n_i > 1. \end{aligned} \quad (1)$$

TTFT captures the queuing and prefill delay before a response begins, whereas TPOT captures the average interval between consecutive tokens after generation starts. Each request is associated with a pair of latency objectives, denoted by S_i^{TTFT} and S_i^{TPOT} . Request r_i satisfies its SLOs only if $\text{TTFT}_i \leq S_i^{\text{TTFT}}$ and $\text{TPOT}_i \leq S_i^{\text{TPOT}}$. In this work, *Multi-SLO* refers both to these phase-specific objectives and to their heterogeneous values across requests and applications. Accordingly, serving efficiency is measured by goodput—the rate of requests that satisfy both objectives—rather than by raw request or token throughput alone.

Continuous Batching and Chunked Prefill. In static batching, a newly arrived request cannot join an executing batch until every request in that batch finishes, which wastes GPU capacity as requests complete at different times. Continuous batching instead updates the active batch at iteration boundaries. After each iteration, completed requests leave, unfinished decode requests remain active, and waiting requests

can be admitted into the released capacity. The batch composition can thus evolve throughout generation, improving GPU utilization while allowing the scheduler to reconsider which requests should run at every step.

A prefill request may contain hundreds or thousands of prompt tokens, whereas an active decode request contributes only its next token to an iteration. Scheduling an entire prompt at once can therefore produce a long iteration and delay token generation for the decode requests in the same batch. Chunked prefill divides the prompt into smaller chunks that may be processed over multiple iterations. This enables prefill chunks and decode tokens to be colocated in one batch, while exposing the prefill chunk size as a controllable scheduling variable. Specifically, let $\mathcal{P}_k$ and $\mathcal{D}_k$ denote the prefill and decode requests selected in step k , respectively, and let $c_{i,k}$ be the number of prompt tokens allocated to prefill request r_i . The number of tokens scheduled in this step is

$$T_k = \sum_{r_i \in \mathcal{P}_k} c_{i,k} + |\mathcal{D}_k|, \quad T_k \leq B_{\max}, \quad (2)$$

where each decode request contributes one token and $B_{\max}$ is the engine’s maximum per-step token budget. The resulting queuing delay and intervals between decoding iterations determine whether each request meets its TTFT and TPOT objectives.

B. Multi-SLO Traces for Investigations

We construct three representative LLM inference workloads covering conversational assistance, mathematical reasoning, and software-engineering agents.

Chatbots. We construct chatbot requests from LMSYS-Chat-1M, which contains one million real-world conversations collected from interactions with 25 LLMs. The token length of each recorded response determines its target output length.

Mathematical Reasoning. We use all 8,792 questions from the training and test splits of GSM8K. Each word problem serves as the input prompt, while the token length of its reference solution determines the target output length.

Software Agents. We use the software-engineering subset of AgentWorldBench and retain one request from each of 99 distinct trajectories. For each request, we generate 1,024 variants by prepending a distinct synthetic prefix while preserving the original agent action and target environment observation.

From the three request pools, we randomly sample 10,000 chatbot, 8,000 mathematical-reasoning, and 4,000 software-agent requests to form the replay traces.

Arrivals and SLOs. We associate requests in order with timestamps from failure-free entries in BurstGPT and replay each workload according to these timestamps, thereby preserving the trace’s temporal arrival pattern. For each workload, we first determine the sending rate λ_{90} at which vLLM achieves a 90% SLO attainment rate. We then set the sending rate to $1.2\lambda_{90}$ to emulate a workload under pressure.

For each request, we execute it alone on an otherwise idle vLLM deployment and record its measured TTFT and TPOT.

TABLE II
EXPERIMENTAL SETUP.

Specifications	
Hardware	2× Intel Xeon Gold 6530 CPUs, 64 physical cores 128 logical threads, 256 GiB host RAM 4× RTX 6000 Ada GPUs (48 GB each)
Software	meta-llama/Llama-3.1-8B ,Qwen/Qwen3-14B, Qwen/Qwen3-32B ; LMSYS-Chat-1M, GSM8K, AgentWorldBench-SWE, BurstGPT

TABLE III
SLO ATTAINMENT OF vLLM ON THE MATH WORKLOAD.

SLO Class	TTFT (%)	TPOT (%)	Joint (%)
Overall	75.80	77.68	53.48
TTFT-prioritized	48.89	100.00	48.89
TPOT-prioritized	100.00	12.00	12.00
Balanced	57.39	100.00	57.39
Relaxed	100.00	100.00	100.00

These isolated measurements serve as request-specific latency baselines. We derive the two SLOs of request r_i as

$$\begin{aligned} S_i^{\text{TTFT}} &= c_i^{\text{TTFT}} \widehat{\text{TTFT}}_i, \\ S_i^{\text{TPOT}} &= c_i^{\text{TPOT}} \widehat{\text{TPOT}}_i, \end{aligned} \quad (3)$$

where $\widehat{\text{TTFT}}_i$ and $\widehat{\text{TPOT}}_i$ are the isolated-run measurements, and c_i^{TTFT} and c_i^{TPOT} are the coefficients assigned to the request’s SLO class. Using a fixed random seed, we assign requests equally among four SLO classes: TTFT-prioritized, TPOT-prioritized, balanced, and relaxed. For Qwen3-14B, the TTFT/TPOT coefficient pairs for these four classes are (20, 2.5), (50, 1.5), (25, 2), and (50, 3), respectively. All compared systems use the same request sequence, arrival schedule, and SLO targets under each experimental condition.

C. Low Multi-SLO Serving Performance

We use vLLM’s default FCFS scheduler as a representative serving baseline. It supports continuous batching and chunked prefill, but schedules requests without differentiating their TTFT and TPOT objectives. We run Qwen3-14B with TP = 2 on two GPUs using the configuration in Table II. We use the math workload as a representative case and replay it at its stress load of $1.2\lambda_{90}$.

Table III summarizes the SLO attainment rates for the math workload. Under FCFS, the overall TTFT and TPOT attainment rates are 75.80% and 77.68%, respectively, but only 53.48% of requests satisfy both objectives. The gap between individual and joint attainment shows that satisfying either latency objective alone does not provide effective Multi-SLO service.

The class-level results further reveal distinct failure modes. TTFT-prioritized requests achieve 48.89% TTFT attainment and 100.00% TPOT attainment, whereas TPOT-prioritized requests achieve 100.00% and 12.00%, respectively. Balanced and relaxed requests attain both objectives for 57.39% and 100.00% of their requests, respectively. The contrasting class-

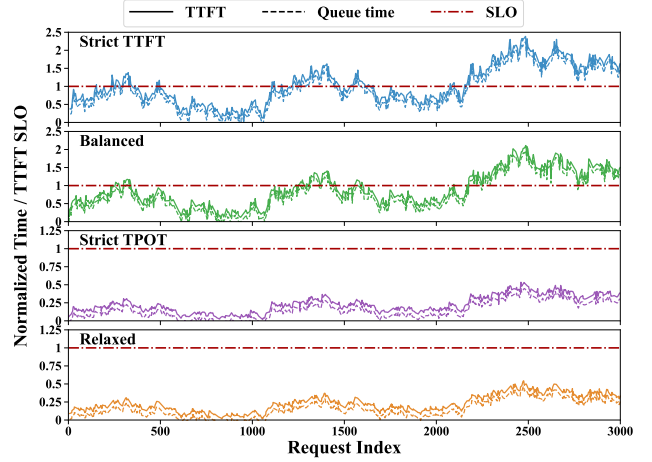


Fig. 2. Normalized TTFT and queueing time under FCFS.

level results indicate that the low joint attainment cannot be attributed to a single global latency bottleneck.

Together, these observations expose a mismatch between throughput-oriented scheduling and request-specific latency objectives. FCFS fixes the service order according to request arrival and does not account for the remaining TTFT or TPOT slack when admitting prefill chunks or advancing decode requests. Consequently, a scheduling decision that preserves prompt responsiveness may disrupt token delivery, while one that protects decode progress may increase the waiting time of queued prefills. The serving engine can therefore retain sufficient capacity to satisfy 100.00% of the relaxed requests, yet fail to direct that capacity toward the phase-specific objectives of other requests. These observations motivate a closer examination of how request ordering and per-step token allocation jointly affect Multi-SLO attainment.

D. Diving into Underlying Reasons

Our investigation identifies two complementary causes of the low Multi-SLO attainment: 1) FCFS distributes queueing delay without considering request-specific TTFT urgency, and 2) a fixed token budget does not provide a reliable latency bound for steps that mix prefill and decode tokens.

1) SLO-unaware temporal scheduling. The first-token latency of request r_i can be decomposed as

$$\text{TTFT}_i = W_i^{\text{queue}} + L_i^{\text{prefill}}, \quad (4)$$

where W_i^{queue} is the delay from arrival until the request is first admitted for prefill, and L_i^{prefill} is the remaining latency until its first token is produced. For a chunked prefill, L_i^{prefill} includes the execution and intervening scheduling delays of all prompt chunks after admission.

Fig. 2 plots TTFT and queueing time for the first 3,001 math requests, normalized by each request’s TTFT SLO. Solid and dashed lines denote TTFT and queueing time, respectively; the dash-dotted line marks the SLO boundary. The Strict TTFT and Strict TPOT panels correspond to the TTFT-prioritized and TPOT-prioritized classes, respectively. The normalized

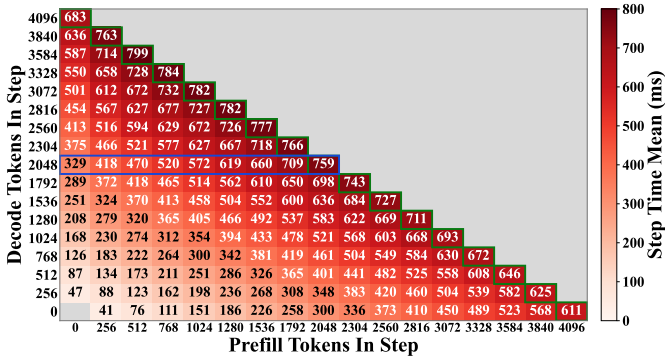


Fig. 3. Mean step time with different prefill and decode token counts.

queueing-time curve closely follows the TTFT curve in all four SLO classes, indicating that queueing dominates the first-token latency. For TTFT-prioritized and balanced requests, the queueing delay frequently consumes most, or even all, of the TTFT budget before prefill completes. TPOT-prioritized and relaxed requests remain below the boundary because their looser TTFT objectives provide substantially more slack. The resulting TTFT violations therefore arise primarily from how waiting time is distributed among requests, rather than from a uniform shortage of prefill compute capacity.

FCFS creates this imbalance because it admits requests according to arrival order, regardless of their remaining TTFT slack. A TTFT-prioritized request may wait behind earlier requests whose looser objectives could tolerate additional delay; the urgent request exhausts its latency budget in the queue while those ahead of it still finish within their SLOs. The temporal scheduling problem is thus to determine *which request* should receive prefill service first. Accounting for the remaining TTFT slack can redistribute queueing delay toward requests that can tolerate it, without changing the total amount of prefill work.

2) Workload-unaware spatial scheduling. The second failure mode arises after requests enter the running batch. For an active decode request, the interval before its next token is determined by the duration of the engine step in which that token is scheduled. Reordering waiting requests alone cannot protect TPOT because a decode token may still share a step with a large prefill chunk.

To quantify this interference, we vary the numbers of prefill tokens p and decode tokens d scheduled in the same step and measure the mean step time. Each configuration is measured on Qwen3-14B with $TP = 2$ over 50 timed steps (5 rounds $\times$ 10 measurements); gray cells exceed the maximum per-step token budget. As highlighted by the blue outline in Fig. 3, when $d = 2,048$, increasing p from 0 to 2,048 increases the mean step time from 329 ms to 759 ms. Every active decode request must wait for the longer step to complete before receiving its next token, directly increasing its TPOT.

More importantly, the total token count alone does not determine step time. Along the green boundary where $p + d = 4,096$, the pure-prefill and pure-decode configurations take

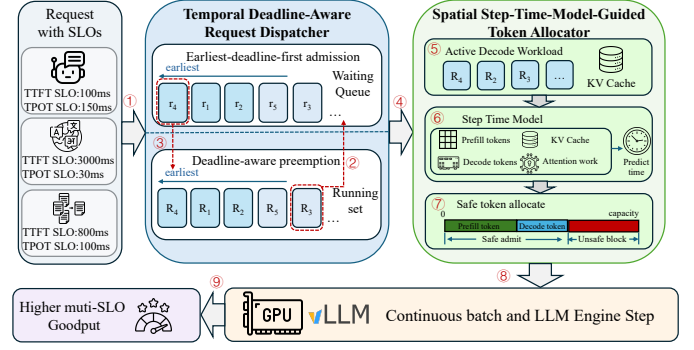


Fig. 4. Design overview of TANGO.

611 ms and 683 ms, respectively, while mixed configurations reach up to 799 ms. Thus, the same total token budget produces about a $1.3\times$ difference in latency. This nonlinearity reflects the distinct execution characteristics of compute-intensive prefill and memory-intensive decode, together with their interference when colocated in one iteration. Consequently, enforcing only $p + d \leq B_{\max}$ bounds the amount of scheduled work but cannot reliably bound the wall-clock latency experienced by active decode requests.

The spatial scheduling problem is therefore to determine *how many prefill tokens* can safely share the next step with the current decode workload. This requires an estimate of mixed-step latency and a mechanism that converts the remaining TPOT slack of active requests into a safe prefill allocation. Temporal request ordering and spatial token allocation address different failure modes: optimizing either dimension alone leaves the other source of SLO violations unresolved. These observations motivate their joint design in TANGO.

IV. TANGO METHODOLOGY

Fig. 4 shows the design overview of TANGO, which comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. At each iteration boundary, the two components coordinate request execution priority and per-step token allocation according to heterogeneous TTFT and TPOT objectives. They take each request's SLOs, inference phase, and execution progress as input and produce a mixed batch of prefill chunks and decode tokens.

The temporal scheduling challenge is to compare request urgency across inference phases. A request with an unfinished prefill must produce its first token within its TTFT budget, whereas a decoding request must sustain token-generation progress within its TPOT budget. The dispatcher maps these phase-specific requirements to comparable deadlines and updates request priorities as inference progresses. These priorities guide the admission of waiting requests and the preemption of less urgent ongoing requests under resource contention (Section IV-A).

The spatial scheduling challenge is to advance urgent prefills while limiting their interference with decoding requests. As shown in Section III-D, the total token count alone cannot reliably determine mixed-step execution time. The allocator

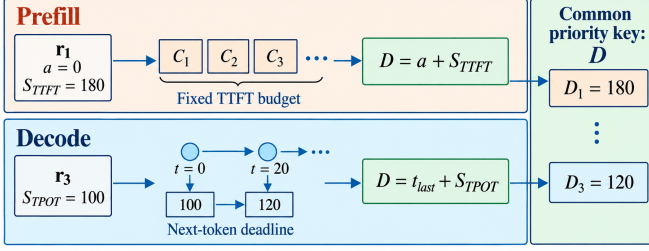


Fig. 5. Phase-aware deadlines for request prioritization (times in ms).

therefore uses a step-time model to predict the duration of candidate batches from their prefill/decode composition and context-dependent attention work. It converts the remaining deadline slack of selected decode requests into a prefill token budget and distributes that budget according to the dispatcher’s priorities (Section IV-B).

Specifically, TANGO operates as follows. First, the scheduler collects the running requests’ execution state, and the allocator derives a prefill token budget using the step-time model. Second, the dispatcher orders waiting requests by their phase-aware deadlines for admission alongside ongoing requests. If a waiting request cannot obtain KV-cache blocks, the dispatcher considers preempting the running request with the latest deadline, but proceeds only when the waiting request has an earlier deadline. The allocator applies the remaining token and time budgets to the prefill chunks of ongoing and newly admitted requests. Finally, the engine executes the resulting batch and returns token timestamps and request progress to update the next scheduling decision. Unfinished requests remain eligible for reconsideration at the next iteration boundary.

The example in Fig. 4 illustrates how the two components cooperate. An urgent waiting request r_4 receives earlier admission, while a less urgent ongoing request can be preempted to release capacity. The allocator then determines how much of the admitted request’s prompt can share a step with the active decode workload. When the decode requests have little remaining deadline slack, it limits the prefill chunk size and leaves the remaining prompt tokens for subsequent steps, even if the engine’s maximum token capacity has not been exhausted. Thus, request urgency determines the service order, while predicted step time determines how much work can execute in that order.

A. deadline-aware request dispatcher

Figs. 5 and 6 illustrate the dispatcher, which uses phase-aware deadlines as a common priority signal for request admission and preemption. In this section, we first introduce phase-aware deadline construction (Fig. 5), then explain how these deadlines guide waiting-queue ordering and preemption under KV-cache memory pressure (Fig. 6).

1) *Phase-Aware Request Ordering*: As analyzed in Section III-D, arrival order does not reflect the different latency budgets of waiting requests. Moreover, a request’s active

latency objective changes during inference: before its first output token, it must produce that token within its TTFT budget; afterward, it must make timely progress in decoding. Our key idea is to express these phase-specific objectives as an effective deadline on the same time axis. Let q_i denote the number of output tokens already returned by request r_i . Using the timestamps and SLOs defined in Section III-A, its current deadline is

$$D_i = \begin{cases} a_i + S_i^{\text{TTFT}}, & q_i = 0, \\ t_{i,q_i} + S_i^{\text{TPOT}}, & q_i > 0. \end{cases} \quad (5)$$

The first case bounds the time available before the first token, while the second gives a rolling target for the next token based on the most recent output timestamp, denoted by t_{last} in Fig. 5. Both cases map the active latency objective to the common priority key D , allowing the dispatcher to compare requests across inference phases.

The deadline is maintained according to output progress rather than reset whenever a request is scheduled. During chunked prefill, all chunks share the original TTFT deadline, so both queueing and the execution of successive chunks consume the same latency budget. For example, r_1 in Fig. 5 arrives at 0 ms with a TTFT SLO of 180 ms; its deadline remains 180 ms across all prompt chunks. When the first token is returned, the active objective switches to TPOT. Each subsequent output token updates the timestamp in Eq. 5 and establishes a new next-token deadline. For r_3 with a TPOT SLO of 100 ms, advancing the latest output timestamp from 0 to 20 ms moves its deadline from 100 to 120 ms. This rolling deadline provides a local target for each inter-token interval; meeting these targets is sufficient to satisfy the average-TPOT objective in Eq. 1.

At each iteration boundary, the dispatcher orders waiting requests by increasing D_i and admits them in earliest-deadline-first (EDF) order. Arrival time breaks deadline ties, with request identifiers providing deterministic ordering when both values are equal. The waiting queue contains both newly arrived requests and preempted requests awaiting resumption. A request that has not returned its first token is ranked by its TTFT deadline, while a resumed decode request is ranked by its TPOT deadline, even if recovering its context requires additional prefill work. Admission remains subject to the engine’s sequence and memory limits and the token budget supplied by the allocator. In Fig. 6, r_4 with $D_4 = 60$ ms is considered before r_5 with $D_5 = 200$ ms. This ordering allows requests with later deadlines to absorb more queueing delay, giving earlier service to more urgent requests without changing their total prefill work.

2) *Deadline-Aware Preemption*: During continuous batching, a new or resumed request may fail to obtain the KV-cache blocks required for admission. The dispatcher then uses the same deadline representation to identify a running request that can yield memory. For the current running set $\mathcal{R}_k$, it selects the candidate victim as

$$v = \arg \max_{i: r_i \in \mathcal{R}_k} (D_i, a_i), \quad (6)$$

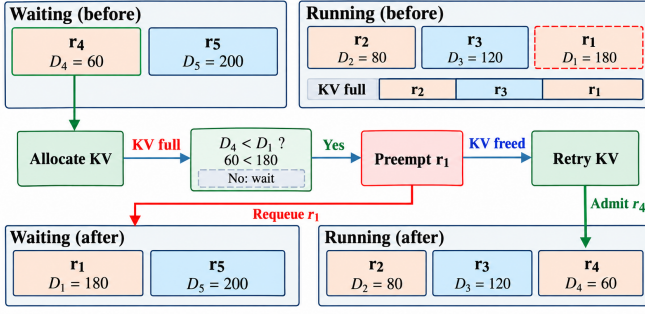


Fig. 6. Deadline-aware admission and preemption (times in ms).

where the tuple is compared lexicographically, so a later-arriving request is selected when deadlines are equal. For waiting request r_j , preemption proceeds only if $D_j < D_v$. Otherwise, r_j remains queued, preserving the running requests whose deadlines are equally or more urgent.

When the comparison permits preemption, the engine reclaims the victim's KV-cache blocks, returns it to the waiting queue, and retries admission of r_j . Preemption occurs at an iteration boundary and preserves the victim's SLOs, arrival time, output-token count, and latest output timestamp. The victim re-enters the queue with the deadline of its current phase: its original TTFT deadline before the first token, or its existing next-token deadline afterward. Thus, preemption does not reset the latency budget, and waiting for resumption continues to consume its remaining slack. In the example at $t = 40$ ms in Fig. 6, KV allocation for r_4 fails, and r_1 has the latest deadline among the running requests. Since $D_4 = 60$ ms is earlier than $D_1 = 180$ ms, the dispatcher preempts r_1 and admits r_4 after retrying allocation. Request r_1 retains its 180 ms deadline and returns to the waiting queue ahead of r_5 , while r_2 and r_3 remain running.

B. step-time-model-guided token allocator

Fig. 7 shows how the allocator determines the amount of prefill work that can share an engine step with the active decode workload. Its key idea is to translate the remaining next-token time budget into a prefill token allocation using a fitted step-time model. In this section, we first introduce the model, then explain how heterogeneous TTFT and TPOT objectives guide budgeting, and finally describe the allocation workflow.

1) *Step-Time Modeling*: As analyzed in Section III-D, batches with the same total token count can have substantially different execution times. The predictor must therefore account for both batch composition and context-dependent attention work. Let p and d denote the numbers of prefill and decode tokens, respectively, K^{dec} the sum of context lengths accessed by decode requests, and A^{pre} the effective attention work of prefill chunks. These four inputs feed the time cost model in Fig. 7, which estimates the step duration as

$$\hat{L}(p, d, K^{\text{dec}}, A^{\text{pre}}) = \alpha + \beta p + \gamma d + \delta K^{\text{dec}} + \epsilon A^{\text{pre}}. \quad (7)$$

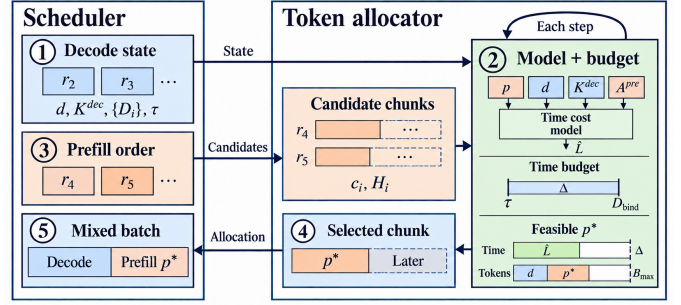


Fig. 7. Execution workflow of the step-time-model-guided token allocator.

Each term approximates a different source of execution cost. The intercept α captures the fixed runtime cost of a step, while βp and γd capture the token-dependent costs of prefill and decode computation. The term δK^{dec} accounts for decode attention's KV-cache reads, which grow with the total history accessed by decode tokens. Finally, ϵA^{pre} accounts for the attention computation performed by prefill chunks.

For chunked prefill, the attention work depends on both the new chunk and its existing prefix. Let $H_{i,k}$ be the already-computed context length available to request r_i before step k . Using the chunk sizes $c_{i,k}$ defined in Eq. 2, we approximate the attention work as

$$A_k^{\text{pre}} = \sum_{r_i \in \mathcal{P}_k} c_{i,k} \left(H_{i,k} + \frac{c_{i,k}}{2} \right). \quad (8)$$

A single fresh prefill thus contributes approximately $p^2/2$, whereas a chunk following a long prefix incurs additional work proportional to that prefix length. Consequently, equal-sized chunks can consume different amounts of the step-time budget. The coefficients are fitted from profiled engine steps for the deployed model and hardware configuration. At runtime, prediction requires only these four workload statistics and a few arithmetic operations.

2) *Joint TTFT/TPOT Budgeting*: At the beginning of step k , the allocator summarizes the active decode workload as $d_k = |\mathcal{D}_k|$ and $K_k^{\text{dec}} = \sum_{r_i \in \mathcal{D}_k} H_{i,k}$. Each decode request contributes its next-token deadline D_i from Eq. 5. The allocator selects a binding deadline from this set using a configurable percentile $\rho_0 \in [0, 1]$. With $\rho_0 = 0$, the earliest deadline bounds the predicted step duration; a larger percentile selects a later deadline and permits more prefill work.

A fixed decode percentile can leave urgent prefills with insufficient service even when increasing prefill work would improve first-token responsiveness. To account for this trade-off, the allocator compares the remaining fractions of the two latency budgets. At scheduling time τ_k , these fractions are

$$u_{i,k}^{\text{TTFT}} = \frac{D_i - \tau_k}{S_i^{\text{TTFT}}}, \quad q_i = 0, \\ u_{i,k}^{\text{TPOT}} = \frac{D_i - \tau_k}{S_i^{\text{TPOT}}}, \quad q_i > 0. \quad (9)$$

The allocator compares the minimum TTFT fraction among requests awaiting their first token in the running set and waiting queue with the minimum TPOT fraction among active decode requests. If TTFT is more urgent, it moves the active percentile ρ_k from ρ_0 toward 1, allowing a later decode deadline to release additional prefill capacity. Otherwise, it retains $\rho_k = \rho_0$. This adaptation gives urgent prefills more service by relaxing the step-time target imposed by earlier decode deadlines.

Let Quantile_{ρ_k} select a deadline at percentile ρ_k from the ordered decode deadlines. Denote the selected deadline by D_k^{bind} , shown as D_{bind} in Fig. 7. The time available for the next step is

$$\begin{aligned} D_k^{\text{bind}} &= \text{Quantile}_{\rho_k}(\{D_i : r_i \in \mathcal{D}_k\}), \\ \Delta_k &= D_k^{\text{bind}} - \tau_k. \end{aligned} \quad (10)$$

The full prediction in Eq. 7 is compared with Δ_k , since fixed and decode-related costs are already included in the model. If no active decode request contributes a TPOT deadline, the allocator uses the remaining engine token capacity without an additional TPOT constraint. If $\Delta_k \leq 0$, it admits no prefill tokens in this step.

For a positive time budget, the allocator seeks the largest feasible prefill allocation:

$$\begin{aligned} p_k^* &= \max_{p \in \mathbb{Z}_{\geq 0}} p, \\ \text{s.t. } p + d_k &\leq B_{\text{max}}, \\ \widehat{L}(p, d_k, K_k^{\text{dec}}, A_k^{\text{pre}}(p)) &\leq \Delta_k. \end{aligned} \quad (11)$$

Here, $A_k^{\text{pre}}(p)$ is computed from the candidate chunks in their scheduling order using Eq. 8. Once the decode workload and candidate order are fixed, only the admitted prefill token count varies, making the budget search one-dimensional. If even a decode-only step exceeds the selected time budget, the allocator sets the additional prefill allocation to zero and allows decode requests to continue making progress. The inequality provides a model-based scheduling target whose accuracy depends on the fitted predictor.

3) *Token Allocation Workflow*: Fig. 7 illustrates the interaction between the scheduler and allocator. At each iteration boundary, the scheduler collects the active decode workload, next-token deadlines, and current time (①). The allocator uses this state to establish the time budget and evaluate the step-time model as prefill candidates are considered (②). The scheduler processes ongoing requests and presents waiting prefill candidates in the order determined by the dispatcher, together with their chunk lengths and existing context lengths (③). For each candidate, the allocator combines these prefill statistics with the decode workload and checks both constraints in Eq. 11 to select a feasible chunk (④). The selected prefill tokens join the active decode tokens to form the mixed batch (⑤).

Running prefill chunks and newly admitted prefills share the same budget. As candidates are allocated, the allocator updates both the cumulative prefill token count and attention work. If the full chunk does not fit, it truncates the chunk to the

largest feasible size; if no positive allocation remains, further prefill work waits for a subsequent step. The dashed segments in Fig. 7 represent this deferred work. Decode tokens remain scheduled subject to the engine’s resource limits.

In the example in Fig. 7, r_2 and r_3 remain active decode requests, while r_4 and r_5 are considered for prefill in that order. The allocator selects a feasible chunk from r_4 ’s prompt, leaving the remaining prefill work for later iterations. Thus, earlier admission does not require an entire prompt to fit in one step. After execution, updated token timestamps and context lengths determine the next allocation.

V. IMPLEMENTATION OF TANGO

TANGO is implemented on top of vLLM 0.17.0 by extending its request admission path and scheduler. The dispatcher and allocator run in the CPU-side scheduling loop and produce per-request token allocations for each engine step. The resulting batch is passed to vLLM through its existing scheduler-executor interface, reusing continuous batching, chunked prefill, PagedAttention-based KV-cache management, tensor parallelism, sampling, and streaming output.

SLO Metadata and Runtime State: We extend request admission to carry two optional SLO fields, TTFT and TPOT, in milliseconds. A missing SLO is treated as an absent constraint, allowing best-effort requests to use the same serving path. Alongside these fields, the scheduler maintains each request’s arrival timestamp, prompt length, computed-token progress, output-token count, and latest output timestamp in host memory. These fields are updated together with vLLM’s request state after each engine step. The first emitted token activates the TPOT objective, and subsequent token timestamps advance the next-token deadline.

Queue and Preemption Hooks: We add queue and preemption hooks to vLLM’s waiting-request admission path. The queue hook applies the phase-aware priority and tie-breaking rules in Section IV-A. On KV-cache allocation failure, the preemption hook selects the running request with the latest deadline and requires the waiting request’s deadline to be strictly earlier. If permitted, vLLM reclaims the victim’s KV blocks, moves it from RUNNING to WAITING, and retries allocation. The victim keeps its SLOs, output progress, and the timestamps used to construct its deadline.

Per-Step Token Allocation: We fit the step-time model coefficients from offline profiling for the deployed model and hardware configuration. At each iteration boundary, the allocator summarizes the active decode workload and collects its next-token deadlines from scheduler-maintained state. It then establishes the time budget using the percentile policy in Section IV-B. As the scheduler processes ongoing prefill requests and admits waiting requests, their chunks share this budget and the engine’s maximum per-step token capacity. The allocator updates the cumulative prefill token count and attention work after each allocation, using the candidate’s chunk length and already-computed context length. A chunk that exceeds either limit is truncated to its largest feasible size; unallocated tokens remain for later steps. The resulting token

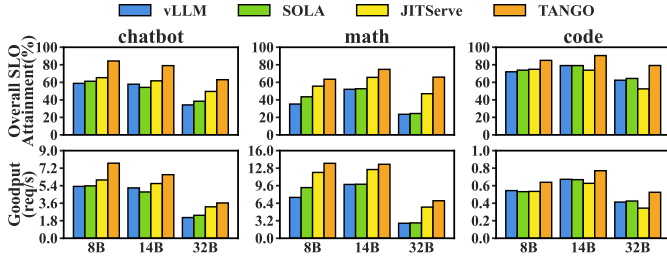


Fig. 8. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe in various scenarios.

allocations are forwarded to the executor through the standard scheduling output.

VI. EVALUATION OF TANGO

In this section, we first evaluate the SLO attainment and goodput of TANGO across different models and multi-SLO workloads. We then investigate the effectiveness of its two scheduling components, validate the step-time model, examine sensitivity to sending rates and SLO settings, evaluate scalability with larger models and GPU deployments, and analyze scheduling overhead.

A. Evaluation Setup

Table II shows our experimental platform. The chatbot, mathematical-reasoning, and software-agent workloads are described in Section III-B. We use Llama-3.1-8B, Qwen3-14B, and Qwen3-32B (8B, 14B, and 32B for short) as base models. Based on the parameter size, we use 1, 2, and 4 NVIDIA RTX 6000 Ada GPUs to deploy them, respectively. Each RTX 6000 Ada has 48 GB of GPU memory. Moreover, we use four SLO classes (TTFT-prioritized, TPOT-prioritized, balanced, and relaxed) for each model.

We use the LLM serving systems vLLM, SOLA, and JITServe as baselines. vLLM employs FCFS admission and a fixed maximum per-step token budget, without differentiating request-specific TTFT and TPOT objectives. SOLA adjusts both request execution order and iteration workload according to request and system states to improve SLO attainment. JITServe estimates uncertain request characteristics and allocates serving bandwidth to individual requests, while grouping compatible requests to improve batching efficiency. Since SOLA’s source code is not publicly available and JITServe uses a different vLLM version, we reimplement SOLA’s scheduling mechanisms and port those of JITServe to vLLM 0.17.0. This allows all systems to be compared on the same vLLM runtime.

Following prior works, we use overall SLO attainment and goodput as performance metrics. Overall SLO attainment is the percentage of all requests that satisfy both their TTFT and TPOT SLOs. Goodput measures the number of such requests completed per second.

B. SLO Attainment and Goodput

We first evaluate TANGO on overall SLO attainment and goodput. Each workload is evaluated on the 8B, 14B, and 32B



Fig. 9. The goodput and overall SLO attainment of TANGO’s variants normalized to TANGO, represented by bars and curves, respectively.

models. For each model–workload combination, we replay the trace at $1.2\lambda_{90}$, where λ_{90} is the sending rate at which vLLM achieves 90% SLO attainment for that combination. All systems use the same request sequence, arrival schedule, and SLO targets. Fig. 8 shows the results for all nine model–workload combinations.

As observed, TANGO achieves the highest overall SLO attainment and goodput in all test cases. Across the nine cases, the average increases in SLO attainment are 23.37%, 21.52%, and 15.48% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are $1.52\times$, $1.47\times$, and $1.22\times$. The most pronounced combined improvement occurs in the Qwen3-32B mathematical-reasoning case, where TANGO increases overall SLO attainment by 42.40%, 41.50%, and 18.90% over vLLM, SOLA, and JITServe, respectively, while the corresponding goodput ratios are $2.50\times$, $2.44\times$, and $1.20\times$.

These improvements are consistent with TANGO’s joint control of request urgency and step duration. The dispatcher prioritizes earlier deadlines for admission and preempts a later-deadline running request when KV-cache allocation fails. The allocator uses predicted step time and active latency slack to bound the prefill work sharing a step with decode requests. This coordination addresses the TTFT violations from queueing and TPOT violations from mixed prefill/decode execution identified in Section III-D.

C. Effectiveness of Temporal Scheduling

In this subsection, we show the performance of TANGO-WOT, a variant of TANGO that disables temporal scheduling. It restores vLLM’s default admission and preemption policies, including FCFS request admission, while retaining the step-time-model-guided token allocator. All other experimental conditions, including the nine model–workload combinations and sending rates, are kept the same as in Section VI-B.

The green bars and curves of Fig. 9 show the goodput and overall SLO attainment of TANGO-WOT normalized to TANGO. As observed, both metrics decrease in all nine cases, with average relative reductions of 20.31% and 20.39%, respectively.

Without phase-aware prioritization, FCFS can leave requests with tight TTFT targets waiting behind earlier requests with more latency slack. The default preemption policy also does not compare active deadlines when deciding which running request should yield KV-cache capacity. Although the retained

allocator regulates the execution time of mixed batches, it cannot by itself give earlier admission to urgent waiting requests. Consequently, controlling step duration alone is insufficient to recover the SLO attainment and goodput of the complete system.

D. Effectiveness of Spatial Scheduling

In this subsection, we show the performance of TANGO-WOS, a variant of TANGO that disables spatial scheduling. It replaces model-guided token allocation with a fixed maximum token budget per step, while retaining the deadline-aware request dispatcher. Thus, request admission and preemption still use phase-aware deadlines, and all other experimental conditions remain unchanged.

The blue bars and curves of Fig. 9 show the goodput and overall SLO attainment of TANGO-WOS normalized to TANGO. Both metrics remain below those of TANGO in every case, with average relative reductions of 13.19% in goodput and 12.66% in SLO attainment.

A fixed token budget does not provide a reliable bound on the duration of a mixed prefill/decode step. As batch composition and context lengths change, the same token count can incur different computation and KV-cache access costs. A long prefill chunk can therefore delay the next tokens of decode requests even when request admission follows deadline order. The model-guided allocator controls this interference by adjusting the prefill allocation to the active decode workload and remaining latency slack. These results show that deadline-aware request ordering and execution-time-aware token allocation make complementary contributions to serving performance.

E. Accuracy of the Step-Time Model

In this subsection, we evaluate the prediction accuracy of the step-time model in Eq. 7. For each deployment configuration, we compare the predicted duration $\hat{L}$ with the measured duration L using its fitted coefficients. The evaluated steps cover decode-only and mixed execution across the three workloads at low and high loads. We evaluate 32,542, 30,988, and 21,952 steps for the 8B, 14B, and 32B models, respectively. We measure prediction error as the absolute percentage error $100|\hat{L} - L|/L$.

Fig. 10 shows predicted versus measured step times and absolute percentage error CDFs (bottom right). As observed, predictions closely match measurements across all three models, with errors concentrated at small values. The mean absolute percentage errors (MAPEs) are 1.83%, 1.62%, and 1.70% for the 8B, 14B, and 32B models, respectively. The fractions of steps with errors below 5% are 90.59%, 93.70%, and 91.38%, respectively. At 10%, the corresponding fractions reach 98.68%, 99.05%, and 98.36%.

F. Sensitivity to the Sending Rate

In this subsection, we investigate the performance of TANGO under different sending rates. The overall comparison in Section VI-B uses $1.2\lambda_{90}$ to evaluate serving performance

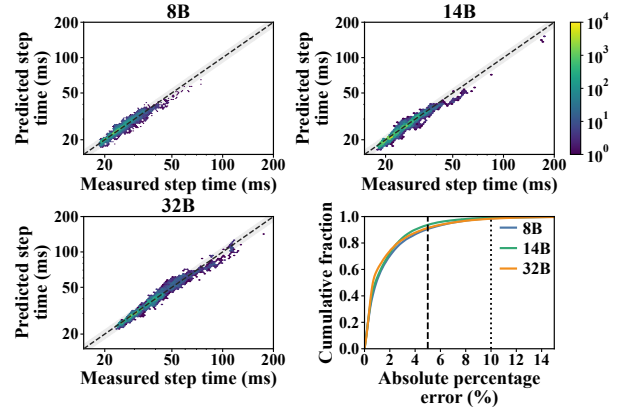


Fig. 10. Step-time prediction accuracy for the three models. Diagonal dashed lines denote perfect predictions; color indicates step counts per hexagon (log scale). The bottom-right panel shows absolute percentage error CDFs.

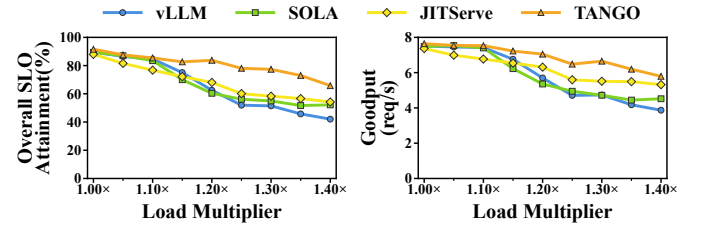


Fig. 11. The goodput and overall SLO attainment of TANGO, vLLM, SOLA, and JITServe under different sending rates for Qwen3-14B on the chatbot workload. The load multiplier is relative to λ_{90} .

under pressure. Here, we vary the load multiplier α from 1.0 to 1.4 in increments of 0.05 and replay the trace at $\alpha\lambda_{90}$. We use Qwen3-14B with the chatbot workload for this experiment. All systems use the same request sequence, arrival schedule, and SLO targets at each sending rate.

Fig. 11 shows the goodput and overall SLO attainment of TANGO and the baselines. At λ_{90} , without additional stress, all four systems achieve high SLO attainment and similar goodput, with TANGO still leading in both metrics. As the sending rate increases, both metrics generally decline, but TANGO maintains higher SLO attainment and goodput, showing a more pronounced advantage under pressure. These results are consistent with TANGO’s coordinated control of request ordering and step duration. As the sending rate increases, more requests compete for execution, making queueing delay and prefill/decode interference harder to control. The dispatcher prioritizes urgent requests for admission, while the allocator adjusts the prefill work in each step according to the predicted duration and active latency slack. This coordination helps TANGO preserve higher SLO attainment and goodput under increasing pressure, although both metrics also decline at higher loads. The results show that TANGO’s performance benefits persist across the evaluated sending rates.

G. Sensitivity to SLO Settings

In this subsection, we evaluate TANGO under different SLO settings. For Qwen3-14B on the chatbot workload, we scale

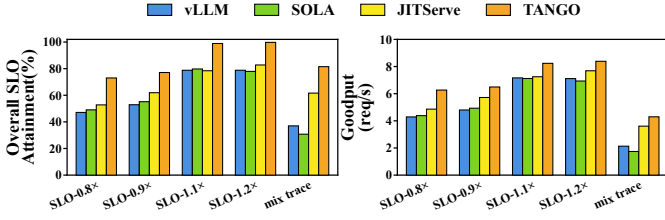


Fig. 12. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe under different SLO settings.

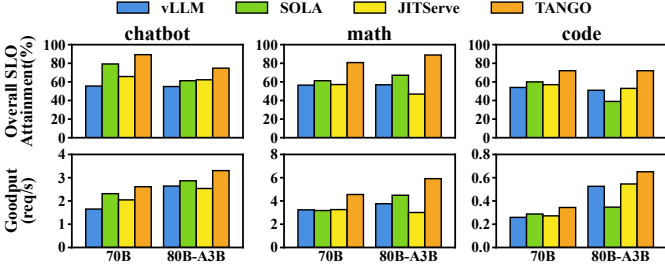


Fig. 13. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe with the 70B and 80B-A3B models on the three workloads.

both TTFT and TPOT targets by $\eta \in \{0.8, 0.9, 1.1, 1.2\}$. We also combine requests from different trace types into a mixed workload. All requests of the same trace type share one fixed TTFT/TPOT SLO configuration, while different types can use different configurations. All systems use identical requests and arrivals within each setting.

Fig. 12 shows that TANGO achieves the highest overall SLO attainment and goodput in all five settings. Across the four SLO scaling factors, the average increases in SLO attainment are 22.80%, 21.72%, and 18.22% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are $1.29\times$, $1.28\times$, and $1.16\times$.

For the mixed trace, TANGO increases SLO attainment by 44.50%, 50.75%, and 19.88% over vLLM, SOLA, and JITServe, respectively, with corresponding goodput ratios of $2.02\times$, $2.47\times$, and $1.19\times$. These results show that deadline-aware request ordering and step-time-guided token allocation remain effective under both SLO scaling and mixed workloads.

H. Scalability with Larger Models and More GPUs

In this subsection, we investigate the scalability of TANGO with larger models and increased GPU resources. We use Llama-3.1-70B and Qwen3-Next-80B-A3B-Instruct (70B and 80B-A3B for short), with the latter using a mixture-of-experts (MoE) architecture. Both models are deployed on eight NVIDIA A800 GPUs with a tensor-parallel degree of 8. We evaluate them on the chatbot, mathematical-reasoning, and software-agent workloads, using the same sending-rate and SLO settings as in Section VI-B.

Fig. 13 shows the overall SLO attainment and goodput of TANGO and the baselines. As observed, TANGO achieves the highest values for both metrics in all six model-workload combinations. The average increases in SLO attainment are

24.79%, 18.29%, and 22.60% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are $1.40\times$, $1.35\times$, and $1.40\times$. The above results demonstrate that TANGO remains effective when scaling to larger dense and MoE models, as well as deployments with more GPUs and greater aggregate GPU memory.

I. Overhead of TANGO

Time Overhead: It mainly comes from deadline maintenance, request ordering, and preemption decisions in the dispatcher, and decode-deadline selection, step-time prediction, and prefill-budget search in the allocator. Both components run on the CPU before model execution and reuse vLLM’s existing executor. The closed-form step-time model requires only a few arithmetic operations for each prediction. For a scheduling step with thousands of candidate requests, the estimated CPU time for deadline maintenance and decode-deadline selection is less than 0.5 ms in total, while step-time prediction and budget search take less than 0.1 ms. Since model execution typically takes tens to hundreds of milliseconds per step, this estimated overhead is usually below 1% of the step latency.

Memory Overhead: TANGO stores additional scheduling state as CPU-side metadata, including request SLOs, arrival and latest-token timestamps, generated-token counts, effective deadlines, and progress information. A compact representation is estimated to require approximately 96 bytes of persistent metadata per request, corresponding to about 240 KB for 2,500 resident requests. Moreover, the allocator stores the fitted step-time coefficients and a few configuration values, with a total footprint below 128 bytes. Its temporary decode-deadline list requires about 16 bytes per active decode request. The per-request state scales linearly with the request count and is negligible compared with the memory occupied by model weights and KV caches.

VII. CONCLUSION

In this paper, we propose TANGO to improve multi-SLO LLM serving through joint temporal and spatial scheduling. Its dispatcher uses phase-aware deadlines to guide request admission and preemption according to heterogeneous TTFT and TPOT requirements. Its allocator uses predicted step time and remaining decode slack to allocate prefill tokens, limiting mixed prefill/decode interference. The evaluation results show that TANGO improves overall SLO attainment by 58.72% and achieves $1.52\times$ the goodput on average compared with state-of-the-art systems.